

SOFTWARE REQUIREMENT SPECIFICATION

Expenditure Management System

- Khushi Yadav (500119315)
- Priyambad Suman (500125300)

INTRODUCTION - Purpose

The purpose of this document is to provide a comprehensive SRS for our Expenditure management System.

This system is designed in a way that it aims to assist users in efficiently tracking and managing their finances.

It allows users to:-

- Track their expenses
- Set monthly budgets
- Generate reports
- Split expenses with friends
- See statistics of their spending

Scope :- This system will allow users to:-

- log in securely using a username and password
- Add and view expenses by date and category
- Set and track a monthly or custom budget.
- Generate and export spending reports
- Search and filter expenses
- split bills with friends and track shared balances.
- View charts and graphs to understand spending better.

Intended Users :-

- Individuals users managing personal expenses

- Students and young professionals
- Anyone who wants an offline personal finance tool.

OVERALL DESCRIPTION

Product Perspective

This is a standalone desktop application. It does not require internet. All features are available offline.

System Features Overview

The system is divided into the following main features:-

- User Authentication
- Budget Management
- Expense Management
- Statistics View
- Report Generation
- Search and Filter
- Split Expenses
- Data Visualization

User Needs

User Need:-

- A secure way to store and view personal financial data
- An easy way to add expenses and track budgets
- Reports to understand spending habits
- A way to share expenses with friends and settle bills.

Assumption and Dependencies

- Users will not share the application with multiple users at once.
- optional features like charts require JFreechart library.

FUNCTIONAL REQUIREMENTS

User Authentication

- Users can log in using a username and password
- Option to register a new account
- Passwords are checked against stored credentials.

Budget Management

- User can set a monthly or custom budget
- After each expense, the remaining budget updates automatically

Expense Management

- Users can add new expenses with:-
 - Amount
 - Data ~~entry~~
 - Category (Food, Travel, Rent, etc.)
 - Description ~~etc~~
- A table shows all added expenses
- Users can create their own categories

Statistics

- View
 - Total expenses
 - Average expense
 - Highest spending category
 - Remaining budget

Report Generation

- Generate a complete expense report
- Option to save/export the report as ~~pdf~~ .csv

Search and Filter

- Search expenses by :-
 - Category
 - Amount range
 - Month or year
- See ~~summaries~~ summaries

Split Expenses

- Add a shared expense with friends/family
- System shows and calculates who owes how much
- (Future upgrade) show borrowing report
- (Future upgrade) Mark balances as paid/settled

Visualization

- View ~~the~~ charts for your expenses

NON FUNCTIONAL REQUIREMENTS

Performance:- High performance

Security:- Used password for security

Usability:- Easy-to-use interface using Java Swing

Portability:- Can run on any system with Java installed

Reliability:- System doesn't crash while adding/editing data

FUTURE ENHANCEMENTS

- Store data in a file or database instead of memory
- Add cloud sync and backup
- Add user notifications and alerts for overspending
- Settlement feature for split expenses

CONCLUSION

This Java Swing application will help users take control of their finances. From adding daily expenses to managing budgets and reports, this project is built with simplicity, usefulness and future scalability in mind.